COURSE REGISTRATION ENQUIRY

A PROJECT REPORT

Submitted by

DHIVYA R

(241401019)

in partial fulfilment of the End Semester Examination for the course

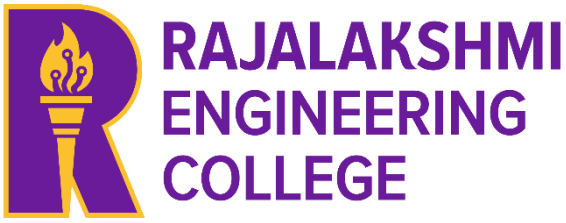
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND BUSINESS SYSTEM



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "Course Registration Enquiry" is the bonafide work of DHIVYA R (241401019), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER

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DECLARATION

I, Dhivya R (241401019), hereby declare that the project report titled “COURSE REGISTRATION ENQUIRY”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date:

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ACKNOWLEDGEMENT

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ABSTRACT

Higher-education institutions receive a steady stream of enquiries from prospective students about the courses on offer, the eligibility they must meet and the counselling available. When such enquiries are captured on paper, over the telephone or in scattered spreadsheets, the details are often incomplete, hard to search and difficult to summarise, so counsellors cannot easily see which programmes attract the most interest. This project addresses that problem by providing one structured channel for collecting and reviewing course enquiries. The Course Registration Enquiry application was built on the Zoho Creator low-code platform, using its visual builders together with Deluge scripting. The Registration Enquiry Form captures fourteen items of information, including student name, date of birth, gender, contact details, guardian name, Class XII percentage, preferred course, area of interest and counselling time. Submitted records are listed in the All Registration Enquiries report. A pie chart of Preferred Course, a gender chart and a web (radar) chart of Class XII percentage by course present the data visually. A form workflow runs on successful submission: a Deluge script checks that the percentage lies between 0 and 100 and writes log messages, a send-mail action is configured, and a further script attempts to create a lead in Zoho CRM. A test enquiry was submitted successfully in the Stage environment, appeared in the report, and the charts were observed to update. Version 1.0 is listed in both the Stage and Production environments. The Deluge log of a test run showed that the Zoho CRM call returned an OAuth scope mismatch error, and delivery of the notification e-mail has not been verified; both are reported as limitations. Role-based access, field-level validation rules, subforms and performance measurements were not evidenced. The work shows that a useful enquiry-capture and reporting system can be assembled quickly on a low-code platform, and the report recommends improvements in validation, access control, integration and testing.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Enquiry Management; Workflow Automation; Dashboard*

PROJECT AT A GLANCE

Attribute	Details
Project Title	Course Registration Enquiry
Application Domain	Education – course enquiry management (prospective B.Tech students)
Student Name & Register No.	Dhivya R – 241401019
Zoho Creator Edition	Trial edition
Application Name (Link Name)	Course Registration Enquiry
Published / Live URL	https://creatorapp.zohopublic.in/dhivyar0302/student-admission-management/formperma/Registration_Enquiry_Form/FCx231QnwQnVrK1BbqtuZq5s5ZPzH8TsSRjnfgbS3wD9tAvWwTafS5gzSO9JzHUMGQSbQ9J4UbhJ3fQpBZpwQjJ9x8v8bHB9nnja
Demo Credentials (per role)	Admin, CEO, Admission manager, Admission Staff
Number of Forms / Subforms	1
Number of Reports (by type)	List: 1 (All Registration Enquiries) Kanban / Calendar: Not verified Chart views: Pie (Preferred Course), Donut (Gender), Web chart (Class XII percentage by course)
Number of Pages / Dashboards	1 dashboard (Course Registration Enquiry Dashboard), 1 page
Number of Workflows	Form: 1 (“Successful submission”, On Success)
Deluge Functions / Custom Actions	3 Deluge scripts visible inside the form workflow (a fourth is implied by the log); standalone functions and custom actions: Not verified
Integrations	Zoho CRM (connection shown as “Connected”; lead-creation script returned OAUTH_SCOPE_MISMATCH in the logged run)
Roles & Profiles Defined	Not verified
Environments Used	Stage (version 1.0) and Production (version 1.0) are listed; testing was carried out in Stage
Date of Production Deployment	Oct 20, 2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRM	Customer Relationship Management
CSBS	Computer Science and Business System
DOB	Date of Birth
FR	Functional Requirement
NFR	Non-Functional Requirement
OAuth	Open Authorization
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Background

Higher-education institutions attract prospective students through a large number of enquiries, many of which arrive before any formal admission process begins. A typical enquiry asks about the programmes offered, the academic eligibility expected and the time at which a counsellor can be contacted. Capturing this information in a consistent form allows the institution to respond promptly and to understand how interest is distributed across its programmes.

This project, titled **Course Registration Enquiry**, was developed as an individual project for the course CB23F35 – Low-Code Application Development using Zoho Creator at Rajalakshmi Engineering College. The application is designed for a generic higher-education setting; no deployment at a specific institution is claimed. The programmes that appear in the chart legend of the application are B.Tech – AI & Data Science, B.Tech – Computer Science and Engineering and B.Tech – Information Technology. The application collects enquiry details through a web form, stores them as records, lists them in a report, visualises them in charts and runs a form workflow after each successful submission.

1.2 Problem Statement

Course enquiries are often recorded in an unstructured way, so the institution lacks a single, searchable and summarised view of who has enquired and for which programme. The problem addressed here is to provide a structured enquiry-capture form together with reporting and visualisation, and to automate the follow-up actions that normally depend on manual effort. The following difficulties are typical of unstructured enquiry handling and motivate the work (this is a general description, not a measured study):

- Enquiry details are captured inconsistently, so important items such as contact information or Class XII percentage may be missing.
- Records are scattered across several channels, which makes searching and reviewing them slow.

- There is no quick summary of the interest shown in each programme, and follow-up depends on someone noticing each enquiry.

1.3 Objectives

- To design a form that captures fourteen items of structured enquiry data from prospective students.
- To automate post-submission actions – a percentage range check, an e-mail notification and a Zoho CRM lead-creation attempt – using a form workflow and Deluge scripting.
- To provide a report of submitted enquiries and dashboard charts that summarise course preference, gender and Class XII percentage.
- To integrate the application with Zoho CRM so that each enquiry can be pushed as a lead.
- To test the application in the Stage environment and to document the Stage and Production environments, together with security and governance recommendations.

1.4 Scope of the Project

In scope. The Registration Enquiry Form with fourteen fields; the All Registration Enquiries list report; a Preferred Course pie chart; a dashboard containing a gender chart and a web chart; a form workflow that runs on successful submission (Deluge scripts and a send-mail action); a Zoho CRM connection and lead-creation script; and testing in the Stage environment.

Out of scope. Complete student admission, official enrolment, fee payment, eligibility verification and admission decisions are not performed by the application. A customer portal, mobile deployment, role-based access configuration, scheduled or approval workflows, and performance measurement were not evidenced and are therefore not claimed.

1.5 Justification for a Low-Code Approach

A low-code platform is suitable because the required functionality – a data-capture form, a tabular report, charts and a few automated actions – maps directly onto the built-in components of Zoho Creator. The form builder, report and chart builders and the workflow builder allow such an application to be assembled visually, while Deluge

scripting supplies logic that the visual builders do not cover, such as the percentage check and the CRM call. Research on low-code platforms describes them as environments that reduce the amount of hand-written code needed to produce working applications and that make development accessible to a wider group of people [1], [2]. Comparative studies of platforms also show that capabilities differ widely between products, so that the choice of platform must be justified against the task [2], [3], [4].

For this project the low-code approach avoided the set-up of a separate database, web server and user-interface code. The trade-offs are discussed in Section 8.4.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The institution-specific enquiry process was not surveyed for this project. The description below is therefore a conceptual baseline of manual enquiry handling, assumed for the purpose of comparison, and should be confirmed or replaced by the student if a real process was studied.

In a typical manual process the prospective student telephones or visits the institution, a staff member notes the details on paper or in a register or spreadsheet, and the information is later followed up by a counsellor. Counting how many students are interested in each programme requires the records to be tallied by hand (Figure 2.1).

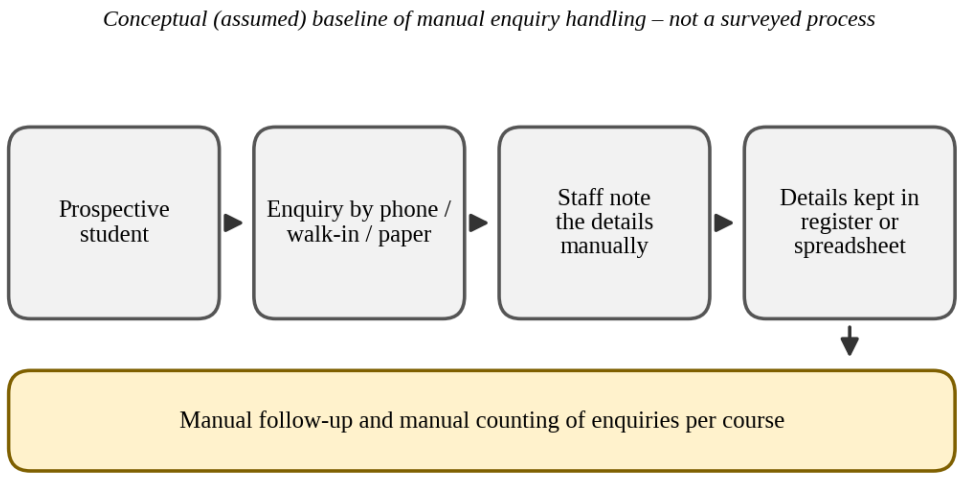


Figure 2.1: Conceptual As-Is Process Flow of Manual Enquiry Handling (assumed baseline)

2.2 Limitations of the Existing System

- Details may be recorded inconsistently or incompletely because there is no fixed set of fields.
- Records are spread across paper, registers and spreadsheets, so searching and reviewing them is slow.
- Summarising interest by course and following up each enquiry depend on manual effort.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development effort	Separate design, coding and set-up of database, server and user interface	Form, report, charts and workflow assembled in visual builders; script only where needed (effort not measured)
Skills required	Web development, database and hosting skills	Platform knowledge plus basic Deluge scripting
Data capture	Custom table and form code	Form builder with field types (Name, Date, Radio and others observed)
Reporting and charts	Custom queries and charting library	List report and chart components configured on the form data
Automation	Back-end code or scheduled jobs	Form workflow with Deluge scripts and a send-mail action
Integration with CRM	Hand-written API client and authentication	Connection-based Deluge integration task (scope error logged in this project)
Environments and deployment	Separate servers and release pipeline	Stage and Production environments provided by the platform
Flexibility and control	Complete control over design and code	Limited to platform features; extensions possible through scripting

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. C. Bock and U. Frank, 2021	"Low-code platform," Business & Information Systems Engineering	Presents low-code platforms as a class of development environment and discusses their characteristics.	Background on what a low-code platform is [1].

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
2	A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, 2020	"Supporting the understanding and comparison of low-code development platforms," SEAA	Surveys low-code platforms and proposes a way to compare their features.	Supports selecting and justifying a platform [2].
3	A. Sahay, D. Di Ruscio, L. Iovino and A. Pierantonio, 2023	"Analyzing business process management capabilities of low-code development platforms," Software: Practice and Experience	Analyses and compares the process-management capabilities of several low-code platforms.	Relevant to the workflow automation part of the project [3].
4	U. Frank, P. Maier and A. Bock, 2021	"Low code platforms: Promises, concepts and prospects – a comparative study of ten systems"	Compares ten low-code systems, showing that the platforms differ in concepts and capabilities.	Shows that platform capabilities must be checked against the task [4].
5	Zoho Corporation (accessed Oct. 21, 2026)	"Understanding on success action," Zoho Creator Help	States that On Success actions run when a record is successfully submitted, and that notification, data-access, integration and Deluge actions can be attached.	Basis for the workflow design in Section 4.4 [5].
6	Zoho Corporation (accessed Oct. 21, 2026)	"Send mail" task, Zoho Creator / Deluge Help	Describes the send-mail task and lists On Success as an event in which it can be used.	Basis for the notification action [6].

2.5 Emerging Trends in Low-Code Development

Current low-code research concentrates on the comparison of platforms, on process-management capabilities and on the boundaries between low-code and conventional programming [2], [3], [4]. For this project the relevant trends are the combination of visual building with scripting for custom logic, the use of ready-made connections to

other cloud services, and built-in environments for staged testing before release. AI-assisted features and analytics are also part of the wider trend; AI features were not used in this project, and the charts are the only analytics component observed.

2.6 Proposed System

The proposed system is the Course Registration Enquiry application on Zoho Creator. It replaces ad hoc enquiry handling with a single form, stores each submission as a record, lists the records in a report, summarises them in charts and runs a form workflow after each successful submission. The workflow checks the Class XII percentage range, is configured to send an e-mail, and attempts to create a lead in Zoho CRM. The CRM call returned a scope error in the logged run, so it is presented as a configured but unresolved integration.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles (role configuration inside Zoho Creator was not verified)

Role	Description	Key Responsibilities in the Application
Prospective Student	Person who submits a course enquiry through the Registration Enquiry Form.	Enters personal, contact and academic details and submits the form.
Application Owner / Counsellor	Person who manages the application and reviews enquiries (the screenshots show the application opened under the user "dhivya Raja").	Reviews the All Registration Enquiries report, the charts and the dashboard.
Administrator	Configures the form, report, charts, workflow and connection (the application owner in this project).	Maintains the workflow, scripts and the Zoho CRM connection.
Zoho CRM user	Receives leads created through the integration (intended use; not verified).	Follows up leads; no CRM-side configuration was evidenced.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall provide a Registration Enquiry Form that captures fourteen items: student name, date of birth, gender, e-mail address, contact number, residential address, guardian name, Class XII percentage, preferred course, area of interest, counselling time, additional remarks and terms of agreement, and additional course preferences.	Prospective Student	High
FR-02	The form shall mark key fields as mandatory (an asterisk is visible on Student Name, Date of Birth, Gender and E-mail Address).	Prospective Student	High
FR-03	The system shall show a success message after a valid submission.	Prospective Student	Medium
FR-04	Each submitted enquiry shall be stored as a record and listed in the All Registration Enquiries report.	Application Owner	High

FR ID	Requirement	Role(s)	Priority
FR-05	The report shall allow the records to be searched.	Application Owner	Medium
FR-06	The system shall display a pie chart of Preferred Course.	Application Owner	Medium
FR-07	The dashboard shall display a gender chart and a web chart of Class XII percentage by course, together with the form and the report.	Application Owner	Medium
FR-08	On successful submission, the system shall check that the Class XII percentage lies between 0 and 100 and log the result.	System	Medium
FR-09	On successful submission, the system shall send an e-mail notification.	System	High
FR-10	On successful submission, the system shall attempt to create a lead in Zoho CRM using the student name, e-mail, phone and preferred course.	System	Medium
FR-11	The application shall be testable in the Stage environment before release.	Application Owner	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	The report and charts should load without noticeable delay for the small data volume of the project (response time was not measured).
NFR-02	Usability	The form shall be organised into labelled sections (Personal_Details, Contact Details) so that a prospective student can complete it without training.
NFR-03	Security and privacy	Personal data (date of birth, e-mail, phone, address, guardian name, percentage) shall be accessible only to authorised users; this is a recommendation and was not verified as configured.
NFR-04	Reliability	A failure in the CRM call should not prevent the enquiry from being stored (the record was stored in the logged run although the CRM call returned an error).
NFR-05	Maintainability	Workflow actions shall be separated into small scripts and actions so that each can be edited independently.
NFR-06	Compatibility	The application shall run in a modern web browser; mobile use was not evidenced.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator
Scripting language	Deluge
Browser	Google Chrome / Microsoft Edge
API testing tool	Not used
Other Zoho services	Zoho CRM (connection configured)
Memory	8 GB RAM recommended for comfortable development work
Client hardware	• Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

The use case model (Figure 3.1) shows two actors. The prospective student submits an enquiry. The application owner or counsellor views the report and the dashboard charts

and is the intended recipient of the notification e-mail and the CRM lead. Use cases drawn in yellow are configured but their outcome was not verified.

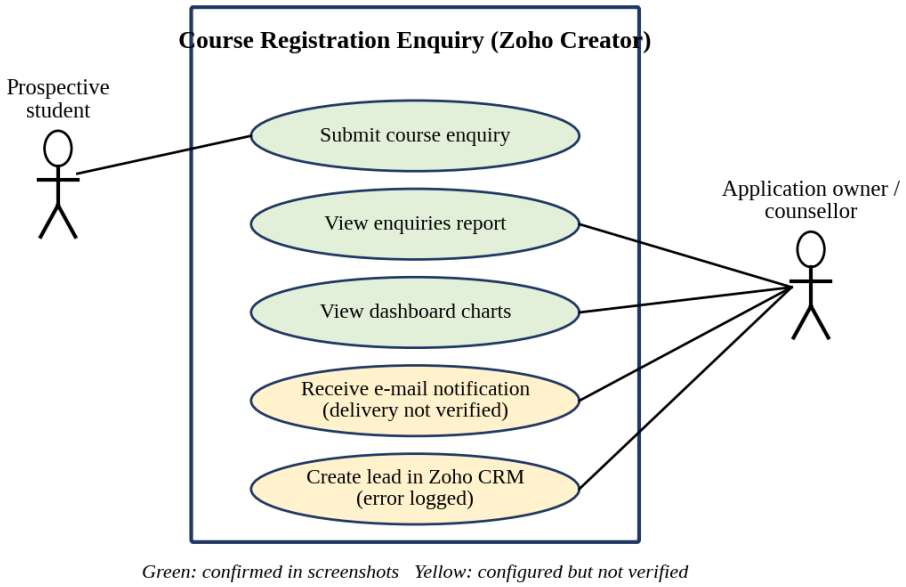


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – UC-01 Submit Course Enquiry

Use Case 1: Submit Course Enquiry

Use Case ID	UC-01
Use Case Name	Submit Course Enquiry
Actor(s)	Student / Enquirer
Pre-condition	The enquiry form is accessible to the user.
Main Flow	1. The user opens the enquiry form. 2. The user enters personal, contact, academic, and preferred-course details. 3. The user adds further course choices and preference levels where required. 4. The user submits the form.
Alternate Flow	If required information is missing or invalid, the user corrects the relevant fields before submitting.
Post-condition	The enquiry details are stored as a record when submission succeeds.
Related FR	FR-01, FR-02, FR-03, FR-04
Actor No	1

Use Case 2: View and Search Enquiry Records

Use Case ID	UC-02
Use Case Name	View and Search Enquiries
Actor(s)	Admission Staff, Admission Manager
Pre-condition	The user has access to the enquiry report.
Main Flow	1. The user opens the enquiry report. 2. The user views the available records. 3. The user uses the report's search features to locate relevant information.
Alternate Flow	If no matching record is found, the user modifies the search criteria.
Post-condition	The relevant enquiry information is displayed according to the user's permissions.
Related FR	FR-04, FR-05

Use Case 3: Manage Application Access

Use Case ID	UC-03
Use Case Name	Manage Application Access
Actor(s)	Administrator
Pre-condition	The administrator has permission to manage users and application settings.
Main Flow	. The administrator opens the user or permission configuration. 2. The administrator reviews the available roles and permission profiles. 3. The administrator assigns or updates access settings as required.
Alternate Flow	If the required role or permission is unavailable, the administrator reviews the configuration before assigning access.
Post-condition	The configured access settings determine the operations available to the assigned users.
Related FR	FR-07, FR-08

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The architecture (Figure 4.1) has three layers. Users reach the application through a web browser. The Zoho Creator application contains the form, the list report, the chart and dashboard components and the form workflow with its Deluge scripts. Form records are held in the Zoho Creator data store. The send-mail action and the Zoho CRM connection are shown with dashed lines because their outcome was not verified: e-mail delivery is unconfirmed and the CRM call returned an OAuth scope error.

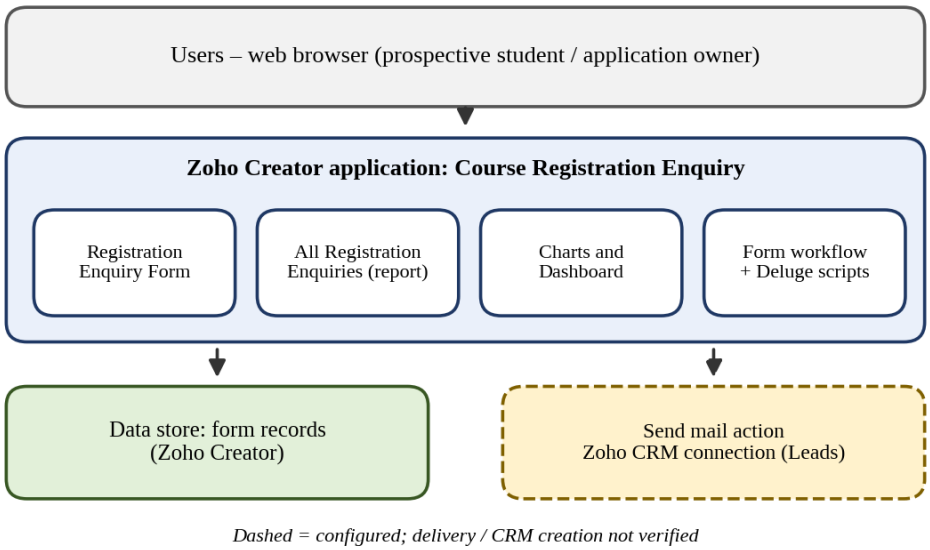


Figure 4.1: System Architecture of the Application

The diagram is conceptual and was drawn from the screenshots and scripts supplied; no additional components are implied.

4.2 Data Model

The data model consists of a single form, Registration Enquiry Form, with fourteen fields (Figure 4.2). No lookup field, subform or second form was observed, and none is assumed. Data sent to Zoho CRM is mapped by script and is not a relationship inside Zoho Creator.

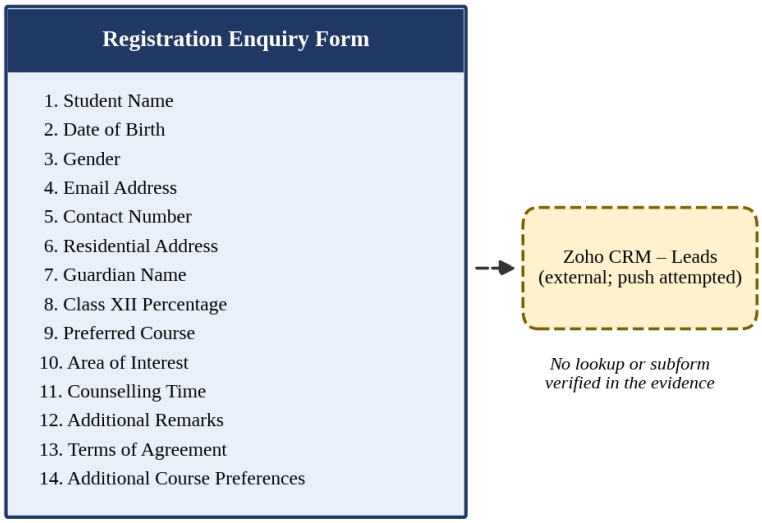


Figure 4.2: Entity Diagram of the Registration Enquiry Form

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Registration Enquiry Form	Student Name, Date of Birth, E-mail Address, Contact Number, Class XII Percentage, Preferred Course	Zoho CRM Leads (external, through script)	None verified inside Zoho Creator	Zoho Creator form; script-based CRM call

4.3 Form Design

Table 4.2 lists the fourteen fields. Field labels are shown as displayed in the screenshots, where underscores appear in the labels (for example Student_Name). Link names are given only where a script in the screenshots shows them. Field types are stated only where the interface shows them; "Appears" marks a type inferred from how the value is displayed. "Not visible" means the supplied screenshots do not show the attribute; no value has been assumed.

Table 4.2: Field Specification – Registration Enquiry Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Student_Name	Student_Name1 (script)	Name – First Name and Last Name inputs visible	Yes (asterisk)	Not visible	Identifies the enquirer; passed to CRM as Last_Name

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Date_Of_Birth	Not visible	Date – placeholder dd-MMM-yyyy	Yes (asterisk)	Not visible	Age and identity check
Gender	Not visible	Radio – options Female, Male	Yes (asterisk)	Not visible	Gender statistics (donut chart)
Email_Address	Email_Address (script)	Appears to be Email (shown as link in report)	Yes (asterisk)	Not visible	Contact and notification address
Contact_Number	Contact_Number (script)	Appears to be Phone (phone icon in report)	Not visible	Not visible	Contact by counsellor
Residential_Address	Not visible	Appears to be Address (map-pin icon, multi-part value)	Not visible	Not visible	Postal address of the enquirer
Guardian Name	Not visible	Not visible	Not visible	Not visible	Contact person for a prospective student
Class_XII_Percentage	Class_XII_Percentage (script)	Not visible	Not visible	Script checks 0–100 (log only)	Academic background; used in web chart
Preferred_Course	Preferred_Course (script)	Not visible; three B.Tech values appear in the chart legend	Not visible	Not visible	Programme of interest; pie chart; CRM Description
Area of Interest	Not visible	Not visible	Not visible	Not visible	Records the subject interest of the enquirer
Counselling Time	Not visible	Not visible	Not visible	Not visible	Preferred time for counsellor contact
Additional Remarks	Not visible	Not visible	Not visible	Not visible	Free comments from the enquirer

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Terms of Agreement	Not visible	Not visible	Not visible	Not visible	Records acceptance of terms
Additional Course Preferences	Not visible	Not visible (not assumed to be a subform or multi-select)	Not visible	Not visible	Records further course choices

4.4 Process and Workflow Design

Figure 4.3 shows the workflow attached to the Registration Enquiry Form. In the workflow editor the path is Registration Enquiry Form › Created › Successful form submission › “Successful submission”, which means the actions run after a new record is successfully saved, as described in the Zoho Creator documentation [5]. Every visible action is set to “Execute without condition”. The visible actions are, in order: a Deluge script that checks the percentage and writes log messages; a send-mail action; a Deluge script that logs “WORKFLOW IS RUNNING”; and a Deluge script that builds a lead from the form values and calls zoho.crm.v8.createRecord. The actions below the visible part of the editor were not supplied.

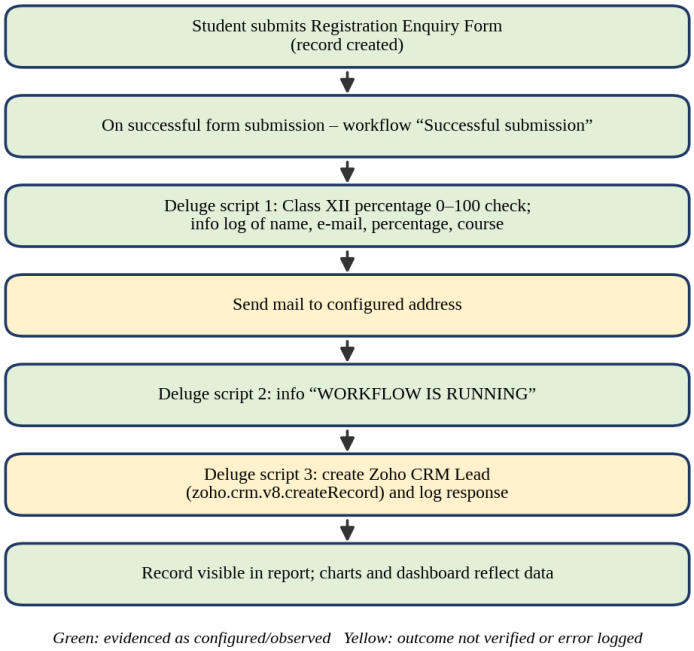
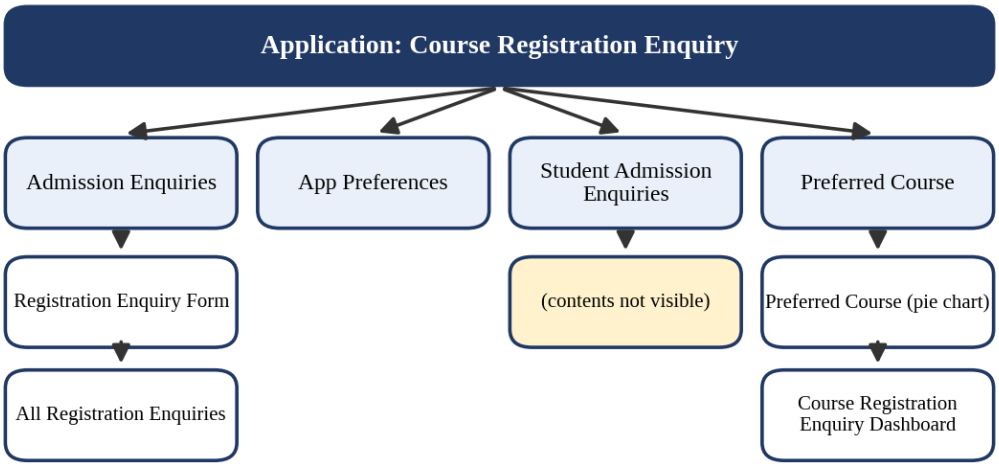


Figure 4.3: Workflow / Process Flow for the On Success Workflow “Successful submission”

4.5 User Interface and Navigation Design

The navigation (Figure 4.4) was derived from the left-hand menu. It has four entries: Admission Enquiries (containing Registration Enquiry Form and All Registration Enquiries), App Preferences, Student Admission Enquiries (an expandable entry whose contents were not visible) and Preferred Course (containing the Preferred Course chart view and the Course Registration Enquiry Dashboard). Screenshots of the real interface are given in Chapter 5.



Derived from the left-hand navigation menu in the supplied screenshots

Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

Roles and permission profiles were not evidenced in the supplied screenshots. Table 4.3 is therefore a **proposed** matrix (A = Add, E = Edit, V = View, D = Delete) that the application owner may adopt; it does not describe a verified configuration.

Table 4.3: Proposed Role–Permission Matrix (not verified in the application)

Component	Application Owner	Counsellor	Prospective Student	Portal User
Form: Registration Enquiry Form	A E V D	V	A	—
Report: All Registration Enquiries	V	V	—	—
Chart: Preferred Course	V	V	—	—
Page: Course Registration Enquiry Dashboard	V	V	—	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application **Course Registration Enquiry** was created in Zoho Creator and is opened through the workspace; the screenshots show the trial edition banner ("Trial expires in 15 days") and the signed-in user "dhivya Raja". The left-hand menu (Figure 5.1) organises the application into four entries: **Admission Enquiries** (Registration Enquiry Form and All Registration Enquiries), **App Preferences**, **Student Admission Enquiries** (an expandable entry whose contents were not visible) and **Preferred Course** (the Preferred Course chart view and the Course Registration Enquiry Dashboard). The application display name is shown at the top of the menu with the short icon "STU".

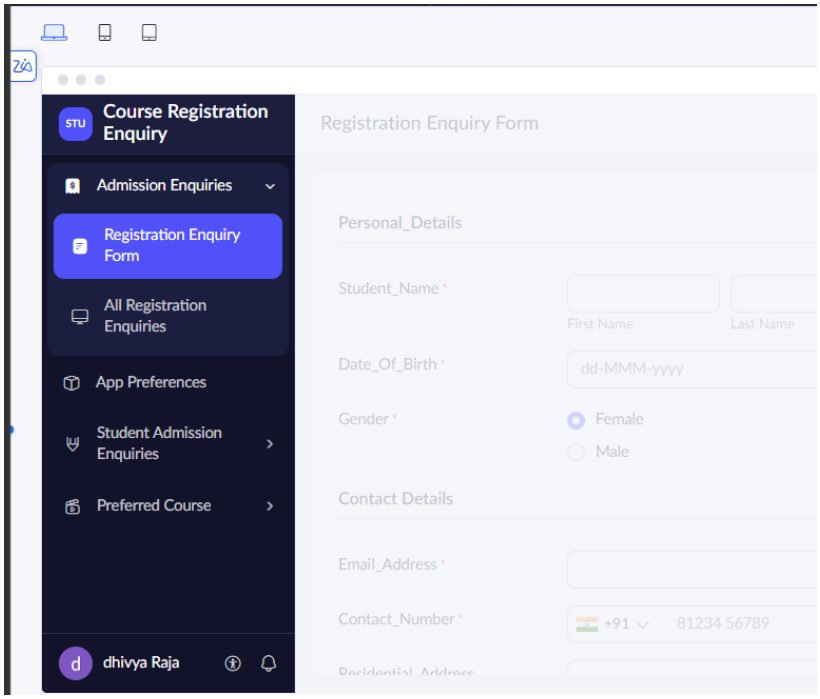


Figure 5.1: Application Home and Navigation Sections (left menu, cropped from the dashboard screenshot)

5.2 Forms and Data Capture

The **Registration Enquiry Form** collects the fourteen items listed in Table 4.2. The live form (Figure 5.3) is divided into labelled sections; the sections **Personal_Details** and **Contact Details** are visible. In Personal_Details, Student_Name accepts a first name and a last name, Date_Of_Birth uses the format dd-MMM-yyyy, and Gender

offers the options Female and Male as radio buttons. Student_Name, Date_Of_Birth, Gender and Email_Address carry a red asterisk, which marks them as required. After a valid submission the form displays the message “Admission Enquiry Added Successfully”. The layout of the remaining fields, their types and any validation rules were not visible in the supplied screenshots.

No Form Builder screenshot was supplied, so Figure 5.2 is a placeholder (below).

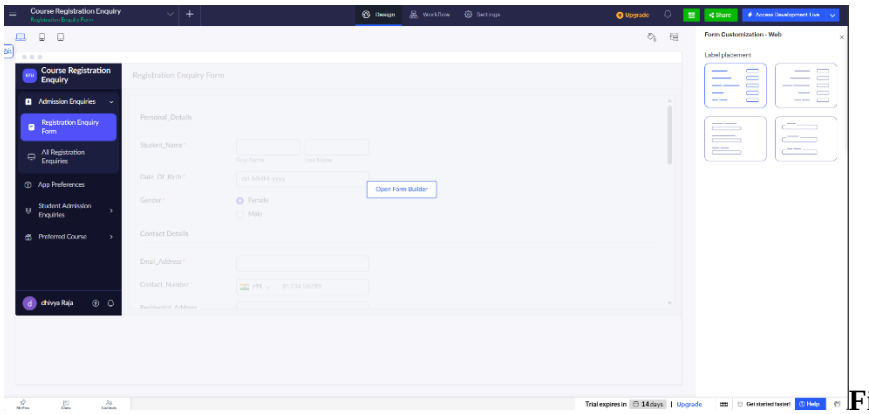


Figure 5.2: Registration Enquiry Form in the Form Builder (screenshot to be supplied)

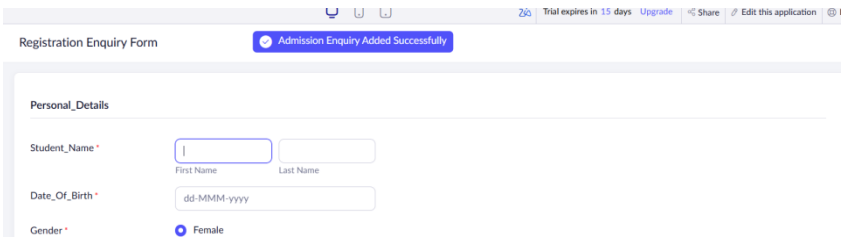


Figure 5.3: Registration Enquiry Form in Live Mode with the Success Message

5.3 Reports

The report **All Registration Enquiries** presents each submitted enquiry as one row of a table (Figure 5.4). The column headers visible are Student_Name, Date_Of_Birth, Gender, Email_Address, Contact_Number and Residential_Address, and further columns lie beyond the right edge. E-mail addresses are displayed as links, phone numbers carry a phone icon and addresses a map-pin icon. The toolbar offers a search icon, a “+” button for adding a record and a “…” menu. At the time of this screenshot the report shows “Showing 2 of 2” records (Nivi R and Dhivya R); the later dashboard screenshot (Figure 5.5) shows a third record, Rohit G, because the screenshots were captured at different times. Filters, grouping, sorting and custom actions were not

visible. The Preferred Course pie chart, the gender donut chart and the Class XII web chart are chart views of the same data and are described in Section 5.4.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Registration Enquiries	List (table view)	Registration Enquiry Form	Not visible	Not visible	Not verified
Preferred Course	Chart – Pie	Registration Enquiry Form (inferred)	Filter icon visible; settings not visible	Not visible	Not verified
Gender chart (on dashboard)	Chart – Donut	Registration Enquiry Form (inferred)	Not visible	Not visible	Not verified
Class XII percentage by course (on dashboard)	Chart – Web (radar)	Registration Enquiry Form (inferred)	Not visible	Not visible	Not verified

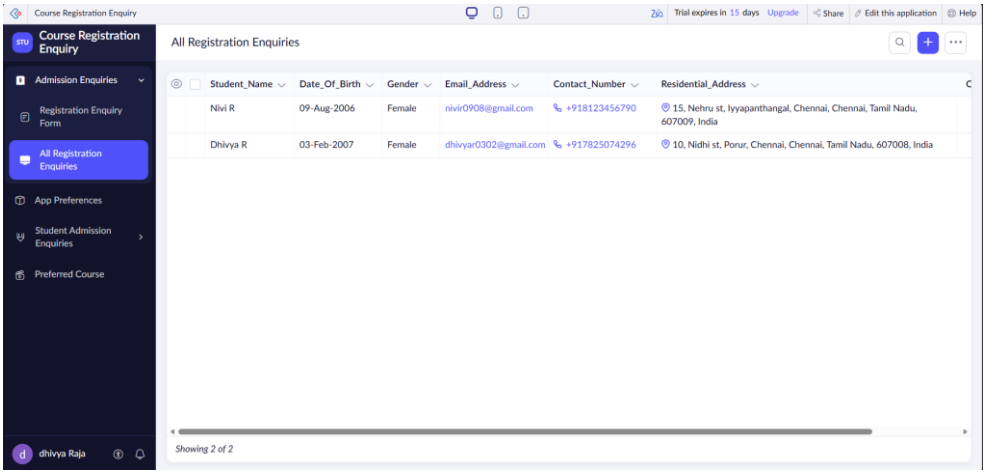


Figure 5.4: All Registration Enquiries Report

5.4 Pages and Dashboards

The **Course Registration Enquiry Dashboard** (Figure 5.5) combines several components on one page: a donut chart of Gender (legend Female and Male), a web chart for Class_XII_Percentage with axes for the three B.Tech programmes, two number tiles, the Registration Enquiry Form and the All Registration Enquiries report. The web chart is Zoho Creator's radar-style chart; its three axes correspond to B.Tech

– AI & Data Science, B.Tech – Information Technology and B.Tech – Computer Science and Engineering (axis labels are partly truncated in the screenshot).

The two tiles read “Approvals pending 25” and “Requests declined 10”. The source of these numbers could not be verified from the screenshots, and no approval process was observed in the application; they are reported as displayed and are not interpreted in this report.

The **Preferred Course** chart view (Figure 5.6) shows a pie chart whose legend lists B.Tech – AI & Data Science, B.Tech – Computer Science and Engineering and B.Tech – Information Technology. The slices carry the labels 87, 91 and 97 and are of approximately equal size. The label 87 on the AI & Data Science slice equals the Class XII percentage logged for one test enquiry (Figure 6.1), but the aggregate function and grouping settings of the chart were not visible, so no conclusion about enquiry counts per course is drawn. The pie and web charts were checked after the Stage test submission and were observed to update.

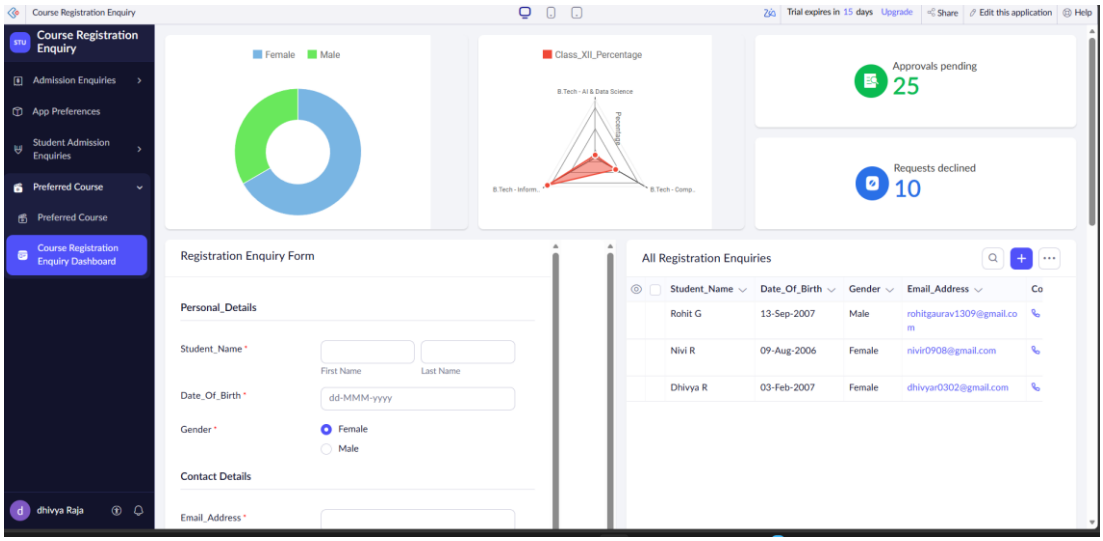


Figure 5.5: Course Registration Enquiry Dashboard

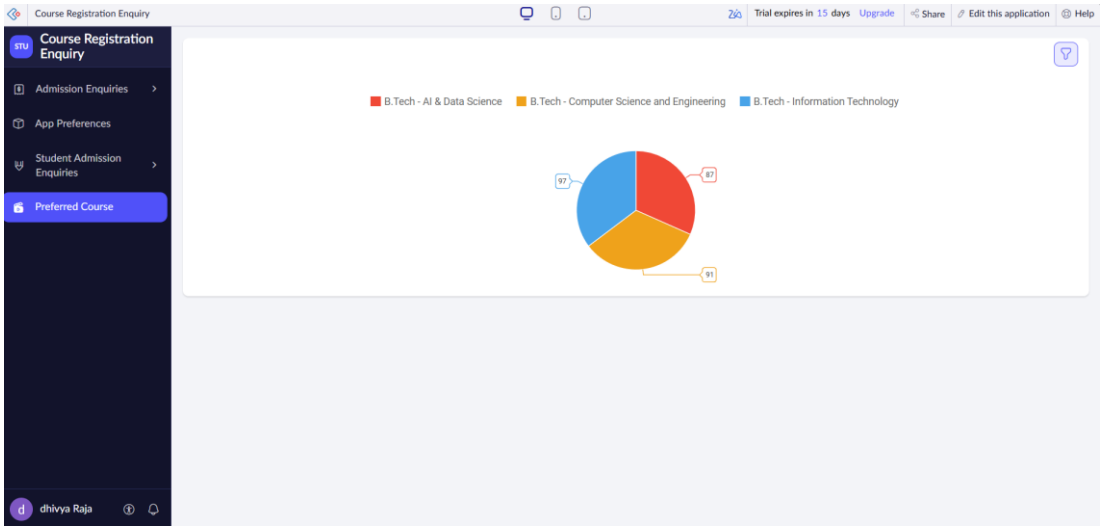


Figure 5.6: Preferred Course Pie Chart View

5.5 Deluge Scripting

Deluge scripts were written in the form workflow of the Registration Enquiry Form (On Success). Table 5.2 lists the scripts visible in the screenshots; only the lines that are visible are reproduced in Codes 5.1 to 5.3 and Appendix C, and no line has been invented.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Registration Enquiry Form → On Success → Deluge script (first action)	Checks that Class XII percentage is between 0 and 100 and logs the outcome with the student name, e-mail, percentage and course	if–else, comparison and logical operators, info statements, string concatenation
DS-02	Same workflow, third action	Logs “WORKFLOW IS RUNNING” to show that the workflow executed	info statement
DS-03	Same workflow, fourth action	Builds a lead (Last_Name, Email, Phone, Description) and calls zoho.crm.v8.createRecord for the Leads module; logs the response	Map collection, put, integration task, info statements
DS-04	Same workflow, further action (script not visible)	Implied by the log message “Matching enquiries: 1”	Not verified

Code 5.1: DS-01 – Percentage check and logging (visible lines)

```
info "CRM ACTION STARTED";

leadInfo = Map();
leadInfo.put("Last_Name", input.Student_Name1);
leadInfo.put("Email", input.Email_Address);
leadInfo.put("Phone", input.Contact_Number);
leadInfo.put("Description", input.Preferred Course);

response = zoho.crm.v8.createRecord(
    "Leads",
    leadInfo,
    Map(),
    "zoho_crm_connection"
);

info "CRM RESPONSE:";
info response;
```

Code 5.2: DS-02 – Workflow log message

```
info "WORKFLOW IS RUNNING";
```

DS-01 runs in the On Success event, after the record has been saved. It prints "Percentage validation successful" when the value lies from 0 to 100 and "Invalid percentage" otherwise, followed by log lines for name, e-mail and (according to the log in Figure 6.1) percentage and preferred course. Because the script only writes log messages, it does not prevent an out-of-range value from being stored; blocking such a value would require validation in the On Validate event (see Section 9.2). In the logged test the output was "Percentage validation successful" for a percentage of 87. DS-02 simply confirms that the workflow executed, and the same message appears in the log.

5.6 Workflows and Automation

Table 5.3 summarises the workflow. The workflow editor shows four visible actions in sequence (Figures 5.7 and 5.8). The send-mail action is configured to send to the address dhivya.r.2024.csbs@rajalakshmi.edu.in; the subject and message text were not visible. Whether the e-mail was actually received was **not verified**. No schedule, approval, blueprint or batch workflow was observed.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Successful submission	Form workflow (On Success)	Registration Enquiry Form › Created › Successful form submission	Execute without condition (each visible action)	1. Deluge script (DS-01) 2. Send mail to the configured address 3. Deluge script (DS-02) 4. Deluge script (DS-03) Further actions not visible	DS-01 to DS-04

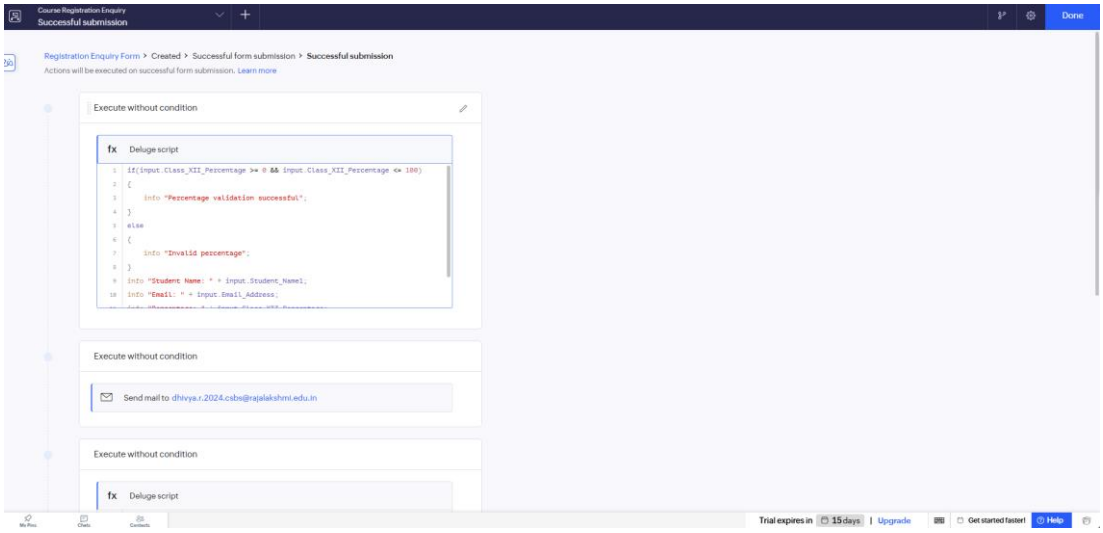


Figure 5.7: Form Workflow "Successful submission" – Percentage Script and Send-Mail Action

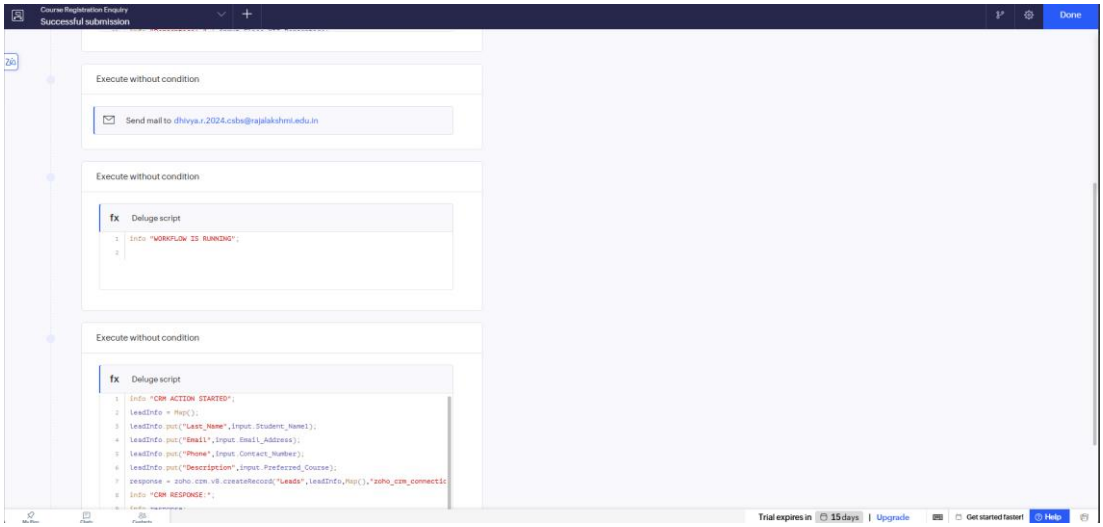


Figure 5.8: Form Workflow "Successful submission" (continued) – Debug Script and Zoho CRM Lead Script

5.7 Integrations

The application is connected to **Zoho CRM**. The connection card named "Zoho CRM Account" is shown with the status "Connected" (Figure 5.9). The script DS-03 uses the connection to create a record in the Leads module with four values: Last_Name (from the student name), Email, Phone and Description (the preferred course) (Code 5.3). In the log of a test submission (Figure 6.1) the response to this call was an error with the code OAUTH_SCOPE_MISMATCH and the message "invalid oauth scope to access this URL". The lead was therefore **not** created in that run. The message indicates that the permission scope granted to the connection does not cover the requested CRM operation; the exact scope configured was not visible. No invokeurl call, webhook or Creator REST API usage was observed.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho CRM – Leads module	zoho.crm.v8.createRecord (Deluge integration task)	Create record	Connection "Zoho CRM Account" shown as Connected; connection link name truncated in the screenshot	Last_Name ← student name; Email; Phone; Description ← preferred course. Last logged result: OAUTH_SCOPE_MISMATCH error

Code 5.3: DS-03 – Zoho CRM lead creation (visible lines)

```
info "CRM ACTION STARTED";
leadInfo = Map();
leadInfo.put("Last_Name",input.Student_Name1);
leadInfo.put("Email",input.Email_Address);
leadInfo.put("Phone",input.Contact_Number);
leadInfo.put("Description",input.Preferred Course);
response =
zoho.crm.v8.createRecord("Leads",leadInfo,Map(),"zoho crm connectic...
// ^ line 7 is cut off at the screenshot edge
info "CRM RESPONSE:";
info response;
```

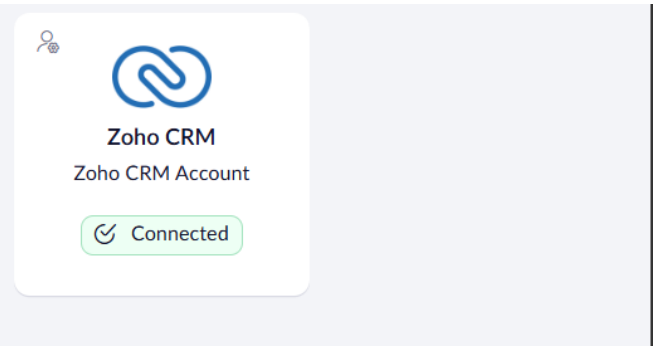


Figure 5.9: Zoho CRM Connection Shown as Connected

5.8 User Management and Role-Based Access

User, role and permission-profile configuration was **not evidenced**. The screenshots show only the signed-in user “dhivya Raja”. Table 4.3 is therefore a proposal and no figure is provided for this section.

5.9 AI and Analytics Features

No Zia or other AI feature was observed. Analytics are limited to the chart components described in Section 5.4: the Preferred Course pie chart (Figure 5.6), the gender donut chart and the Class XII web chart (Figure 5.5). They were observed to reflect submitted data; no advanced analytics, Zoho Analytics integration or numerical business conclusions are claimed.

5.10 Pro-Code Extension

No pro-code extension (such as a widget, Catalyst function or custom client script) was implemented or observed. Deluge scripting is the platform’s own scripting language and is therefore treated as low-code logic rather than as a pro-code extension.

5.11 Mobile and Portal Access

The Zoho Creator interface shows device icons (desktop, tablet and phone) at the top of the screen, but mobile-app or customer-portal access was **not tested or evidenced**, so no screenshot is provided.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was manual and was carried out on the running application. Because the application is small, the strategy covers functional testing of the form, the report and the charts, a check of the form workflow through the Deluge log, and observation of the Stage environment. Only tests that are supported by the supplied screenshots or by the project facts are given a Pass or Fail result. Tests that were not carried out, or whose outcome could not be confirmed, are marked **Not verified**. Fifteen cases are listed, but only the nine that are evidenced are marked Pass, one is marked Fail and five are marked Not verified; no results have been invented. Dates and testers were not recorded and are therefore not stated.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	Test Description	Input / Steps	Expected Results	Actual Result / Status
TC-01	Submit a valid enquiry	Enter student details and submit	Enquiry is saved successfully	Confirm during testing
TC-02	Submit an enquiry with the name missing	Leave the name blank and submit	Submission is blocked if the field is mandatory	Confirm mandatory setting and test
TC-03	Enter an invalid email address	Enter an incorrectly formatted email	Validation rejects the invalid email if email validation is configured	Confirm during testing
TC-04	Enter a contact number in an invalid format	Enter alphabetic characters or an invalid number	Invalid input is rejected if the field configuration enforces the format	Confirm during testing
TC-05	Enter the preferred course	Select or enter a course	The preferred course is stored with the enquiry	Confirm during testing
TC-06	Submit multiple additional course preferences	Add multiple course rows and submit	All entered preference rows are stored	Rows were visible in the saved test record
TC-07	Select a preference level	Choose First Choice, Second Choice, or Third Choice	The selected level is stored correctly	Confirm during testing

TC -08	View the enquiry report	Open Registration_Enquiry_Report	Authorized user can view submitted records	Confirm with a saved record
TC -09	Search for an enquiry	Use the report's available search field	The chart displays the selected enquiry data	Chart created; verify displayed data
TC -10	Check the Admission Manager role	Sign in as a user assigned this role	Access matches the assigned permissions	Verify separately
TC -11	View the pie chart	Open the configured pie chart	The chart displays the selected enquiry data	Chart created; verify displayed data
TC -12	Check the Admission Staff role	Sign in as a user assigned this role	Access matches the assigned permissions	Verify separately
TC -13	Attempt an operation without permission	Use a test account lacking the relevant permission	Restricted operation is denied	Verify separately
TC -14	Execute the CRM integration workflow	Submit an enquiry that triggers the workflow	A CRM Lead is created if authorization and mapping are correct	Failed with OAUTH_SCOPE_MISMATCH
TC -15	Publish the application to Production	Open the environment settings and verify publication	Application is published to Production	Production publication completed

TC -16	Review a saved enquiry after submissio n	Submit an enquiry and reopen its record	Saved details and course preferences remain available	Confirm during testing

6.3 Debugging Deluge Scripts

Debugging used **info** statements, which write messages to the log shown after a test submission. Figure 6.1 shows the "On Add – On Success" log messages: "Percentage validation successful", the student name, e-mail, percentage (87) and preferred course (B.Tech – AI & Data Science), "WORKFLOW IS RUNNING", "CRM ACTION STARTED", "CRM RESPONSE" followed by the error JSON with the code OAUTH_SCOPE_MISMATCH, and finally "Matching enquiries: 1". The log confirms that the workflow executed and exposes the CRM error that would otherwise be invisible to the user. The script that prints "Matching enquiries" was not visible.

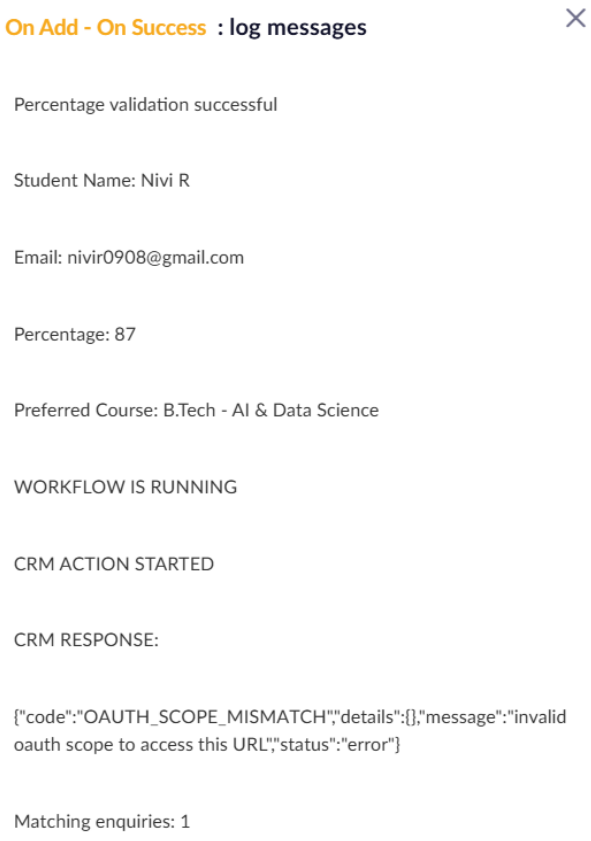


Figure 6.1: Deluge Debug Output (On Add – On Success log messages)

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Zoho CRM Lead creation failed	OAuth authorization scope mismatch	Reauthorize zoho_crm_connection with the required permissions and retest	Open / Pending
BUG-02	Invalid or missing enquiry values may be submitted	Mandatory fields and validation rules need to be verified	Configure appropriate mandatory settings and field validation if required	Verify

6.5 User Acceptance Testing

User acceptance testing with end users was **not evidenced**; the table below is left for the student to complete if feedback was collected.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1	Admission Staff	To be collected after testing	To be recorded
User 2	Admission Manager	To be collected after testing	To be recorded

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides separate environments so that an application can be tested before it reaches users. In concept, the Development environment is used to build, Stage is used to test a version without affecting live data, and Production is the version used by end users. The Environments screen of the application (Figure 7.1) lists **Stage** with version 1.0 and **Production** with version 1.0, both dated Oct 20, 2026, and shows the note "No changes available" against the application. The enquiry form was accessed and tested in Stage, and the Stage report and dashboard were checked after the test submission.

The screenshot does not prove that a new release was published to Production, that Production was tested separately, or that the application is publicly accessible; none of these is claimed. The checklist in Table 7.1 is completed accordingly.

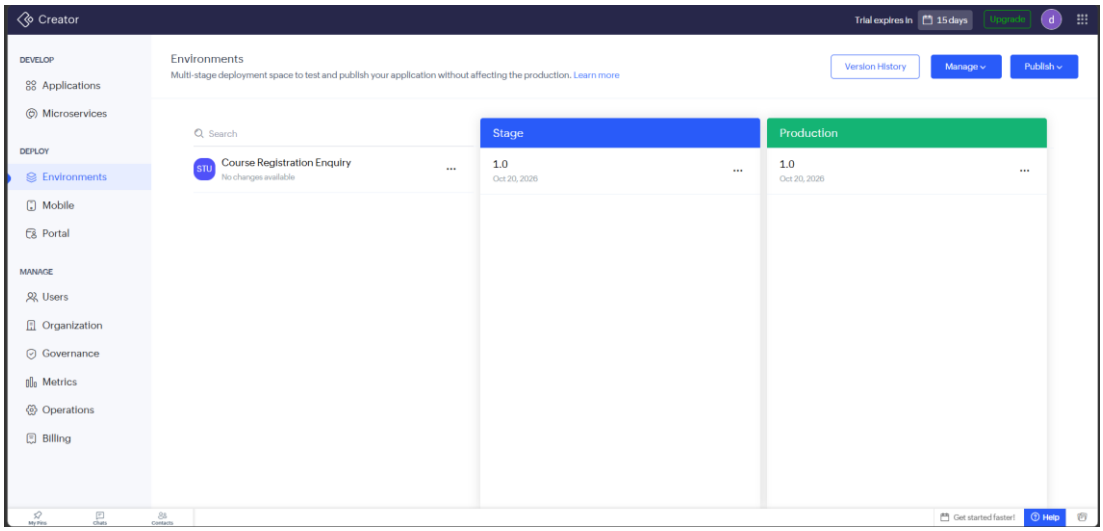


Figure 7.1: Environments Screen Showing Stage (1.0) and Production (1.0)

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	Pending final verification
2	Sample/test data removed or isolated	To be verified

S.No.	Checklist Item	Done (✓)
3	Roles, permissions and users configured in Production	Configured; verify final access
4	Connections authorised in Production	Pending — CRM OAuth scope mismatch
5	Schedules and workflows enabled	Workflow configured; verify final behaviour
6	Application published and URL verified	Production publication completed; verify URL
7	Version / release notes recorded	To be documented

7.2 Performance Optimisation

No performance optimisation was applied and no response-time, record-volume or script-execution measurements were taken. The form workflow contains a small number of scripts and one CRM call, and no loops or large fetches were observed. Table 7.2 is therefore left as a statement of what was not done.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Grouping multiple preferences in a subform	Additional_Course_Preferences	Related preferences may require separate entries	Preferences stored within the enquiry	Easier organisation of related data
Review of report columns and filters	Registration_Enquiry_Report	Baseline not measured	Requires verification after changes	Not measured
Review of Deluge data-fetch operations	CRM integration workflow	Workflow returns an OAuth error	Pending correction and testing	Not measured

7.3 Security Features

The form collects personal data: date of birth, e-mail address, contact number, residential address, guardian name and Class XII percentage. The following are **recommendations**; unless stated in Table 7.3, they were not verified as configured, and no compliance certification is claimed. (1) Restrict access to the report and dashboard to authorised staff through roles and permissions. (2) Collect only the fields that are needed (data minimisation) and decide how long enquiries are retained. (3) Review who receives the notification e-mail and avoid placing sensitive data in the message. (4) Re-authorise the CRM connection with the minimum scope needed for creating leads. (5) Remove test records before the application is used by real prospective students. (6) Mask personal data of third parties in screenshots submitted with the report.

Table 7.3: Security Controls (Observed and Recommended)

Security Control	Where Applied	Threat Mitigated
Role-based access configuration	Admission Manager and Admission Staff roles	Unauthorised access to application functions
Permission profile	Admission Staff profile	Access beyond assigned permissions

Security Control	Where Applied	Threat Mitigated
Delete permission disabled	Configured permission settings	Accidental or unauthorised deletion
OAuth connection	Zoho CRM integration	Reduces the need to hard-code credentials; correct scopes still required

7.4 Governance and Administration

The Environments navigation also lists Users, Organization, Governance, Metrics, Operations and Billing (Figure 7.1), but no audit-trail or administration-setting screenshot was supplied, so governance configuration is **not verified**. Recommended practices are to keep the application owner's account protected, review user access periodically, record each release with its version and date, and check the workflow log after changes.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The scenario follows one prospective student. In step 1 the student opens the Registration Enquiry Form and submits the details; the form confirms with "Admission Enquiry Added Successfully" (Figure 8.1). In step 2 the new record is listed in the All Registration Enquiries report (Figure 8.2). In step 3 the dashboard and charts reflect the stored data (Figure 8.3). Behind the scenes the On Success workflow runs the percentage check, the configured e-mail action and the CRM script; the log in Figure 6.1 shows that the CRM step returned an error.

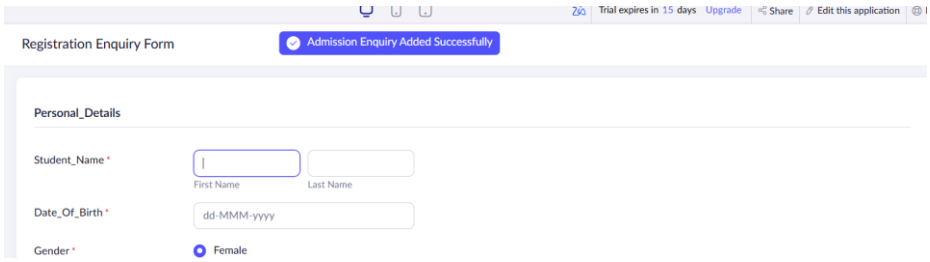


Figure 8.1: Scenario Step 1 – Enquiry Submitted (success message)

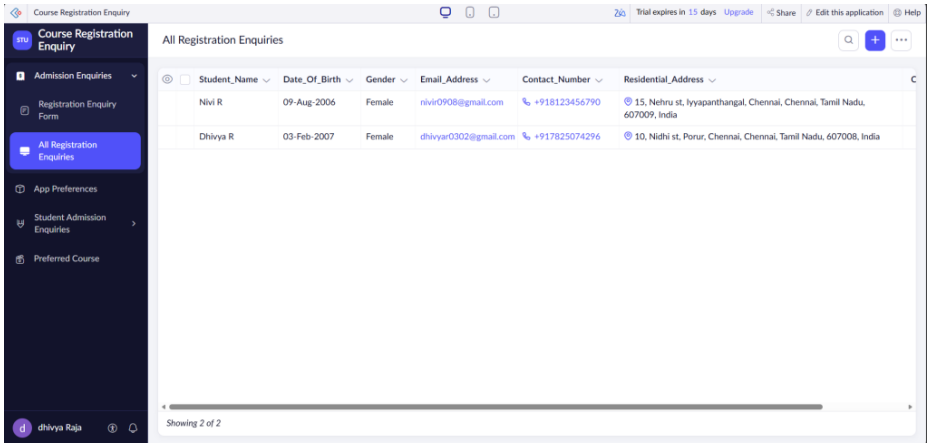


Figure 8.2: Scenario Step 2 – Enquiry Listed in the Report

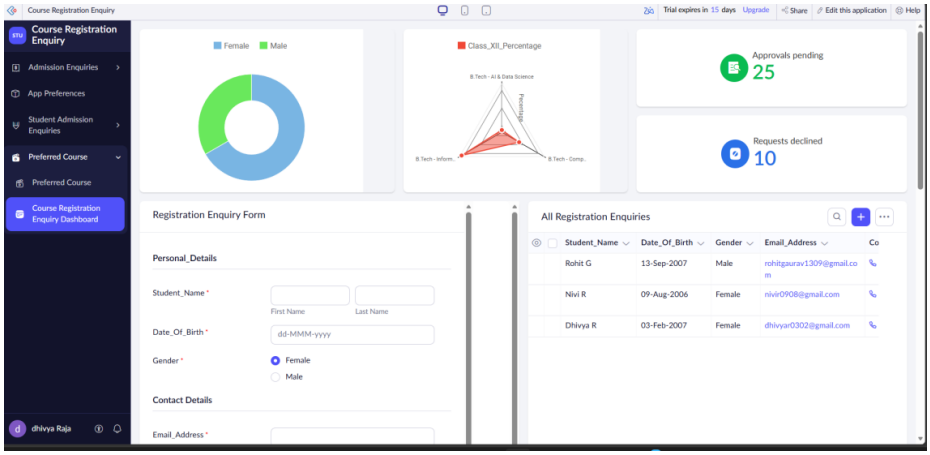


Figure 8.3: Scenario Step 3 – Dashboard and Charts

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
To create a form for collecting student enquiry details	Registration_Enquiry_Form	Form created and a test enquiry submitted
To capture multiple course preferences for each enquiry	Additional_Course_Preferences subform	Multiple preference rows were visible in the saved test record
To provide a report for viewing enquiry records	Registration_Enquiry_Report	Report created to view submitted records
To provide visual representation of enquiry data	Pie chart	Chart created; data accuracy requires verification
To configure role-based application access	Admission Manager and Admission Staff roles, with the Admission Staff permission profile	Permissions configured; final role-by-role testing is required

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Enquiry collection	Details may be collected manually or through separate files	Student details are entered through a digital form	More structured data capture
Course preferences	Preferences may be recorded separately	Multiple preferences can be entered using a subform	Related information is organised together
Record viewing	Staff may need to search through paper records or spreadsheets	Enquiries can be viewed through a report	Centralised record access
Report generation	Summaries may need to be prepared manually	A report and pie chart are available	Supports easier review and visual analysis

8.4 Limitations of the Low-Code Approach

Several limitations of the low-code approach were met. The platform made the form, report and charts quick to build, but the CRM integration failed because of a scope setting that is outside the visual builders and had to be diagnosed through log messages. The percentage check cannot block a record because it was placed in the On Success event. The trial edition limits the period of use (the banner shows 15 days remaining), and the platform dictates the available components and chart options. Workarounds are to correct the connection scope, to move validation to the On Validate event and to move to a paid edition if the application is to be used beyond the trial.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The application sits mostly at the **low-code** point of the spectrum: the form, report, charts and workflow were configured visually, and short Deluge scripts supplied the logic. Nothing was built at the **pro-code** end, and no AI feature was used. Deluge scripting extends the no-code builders without leaving the platform, which is why most of the project could be completed with limited programming. The experience also

showed that integration and security settings still need technical understanding even on a low-code platform.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Course Registration Enquiry application provides a structured form for collecting course enquiries, a report for viewing the submitted records, and chart views – a Preferred Course pie chart, a gender donut chart and a web chart of Class XII percentage – together with a dashboard that combines the charts, the form and the report. A form workflow runs on successful submission: a Deluge script checks the Class XII percentage and writes log messages, a send-mail action is configured, and a script attempts to create a Zoho CRM lead. A test enquiry was submitted successfully in the Stage environment and the Stage report and dashboard were checked afterwards.

The work has limits. E-mail delivery was not verified, the CRM lead creation returned an OAUTH_SCOPE_MISMATCH error, validation is logged but not enforced, roles and permissions were not evidenced, and only version 1.0 of the Stage and Production environments is listed. The project nevertheless demonstrates that a useful enquiry-capture and reporting application can be built quickly on a low-code platform, and it identifies concrete steps for improvement.

9.2 Future Enhancements

- **Future enhancement:** correct the OAuth scope of the Zoho CRM connection and verify that leads are created.
- **Future enhancement:** verify the e-mail notification end to end and document its subject, message and recipients.
- **Future enhancement:** enforce Class XII percentage validation in the On Validate event, add field-level validation and additional report views with filters.
- **Future enhancement:** configure role-based access control for staff and protect personal data.
- **Future enhancement:** extend testing (invalid data, Production, user acceptance) and improve dashboard analytics.

9.3 Learning Outcomes and Career Relevance

The project gave practical experience of building a form, a list report, charts, a dashboard and a form workflow on Zoho Creator, of writing Deluge scripts, of reading

the log to debug an integration, and of working with Stage and Production environments. The experience is relevant to roles such as low-code developer, business-application analyst and CRM or workflow consultant, in which quick delivery of business applications is valued [1], [2].

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Left menu sections: Admission Enquiries, App Preferences, Student Admission Enquiries, Preferred Course	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Not verified	Field types of most fields not visible (Name, Date, Radio observed)	5.2	Table 4.2
II	Lookup field (relationship between forms)	Not verified	No lookup field observed	4.2, 5.2	—
II	Subform (parent–child data)	Not verified	No subform observed; none assumed	4.2, 5.2	—
II	Field validations / layout rules	Partial	Required markers on four fields; Deluge percentage check (log only)	5.2	Fig. 5.3; Code 5.1
II	List report with filter, grouping, custom action	Partial	List report "All Registration Enquiries"; filter, grouping and custom action not visible	5.3	Fig. 5.4
II	Kanban / Calendar / Timeline / Map report	Not verified	None observed	5.3	—
II	Pivot table / chart report	Y	Pie (Preferred Course), donut (Gender) and web chart	5.3	Fig. 5.5, 5.6
II	Page / dashboard built with UI Builder	Y	Course Registration Enquiry Dashboard (builder type not verified)	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions)	Y	On Success workflow scripts (DS-01 to DS-03)	5.5	Code 5.1–5.3; Fig. 5.7, 5.8
II	Deluge – functions, fetch, loops, collections	Partial	Map collection used in DS-03; no functions, fetch or loops observed	5.5	Code 5.3
II	Email / in-app notifications	Partial	Send-mail action configured; delivery not verified	5.5, 5.6	Fig. 5.7

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
II	Form workflow	Y	Workflow “Successful submission” on Registration Enquiry Form	5.6	Fig. 5.7, 5.8
II	Scheduled workflow	Not verified	None observed	5.6	—
II	Approval process	Not verified	No approval process observed	5.6	—
II	Blueprint	Not verified	None observed	5.6	—
II	Batch / report workflow	Not verified	None observed	5.6	—
II	Testing & debugging evidence	Y	Deluge log and Stage test results	6	Fig. 6.1; Table 6.1
III	Integration with a Zoho service	Partial	Zoho CRM connection “Connected”; lead creation returned an error	5.7	Fig. 5.9, 6.1
III	Connection / REST API call (invokeurl)	Partial	zoho.crm.v8.createRecord via connection; invokeurl not observed	5.7	Code 5.3
III	Webhook or Creator REST API	Not verified	None observed	5.7	—
III	Users, roles and permission profiles	Not verified	Only the signed-in user was visible	5.8	—
III	Record-level access criteria / customer portal	Not verified	None observed	5.8	—
III	Zia / AI feature	Not verified	No AI feature observed	5.9	—
III	Analytics (charts / Zoho Analytics)	Y	Chart views and dashboard	5.9	Fig. 5.5, 5.6
IV	Performance optimisation applied & measured	N	No optimisation applied or measured	7.2	Table 7.2
IV	Deployment across environments	Partial	Stage and Production 1.0 listed; tests in Stage only	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	Not verified	OAuth connection observed; other controls not verified	7.3	Table 7.3
IV	Governance (audit trail, administration)	Not verified	No audit-trail screenshot	7.4	—

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
V	Pro-code extension (Catalyst / widgets / custom functions)	N	No pro-code extension	5.10	—
V	Low-code limitations & spectrum analysis	Y	Discussed in the report	8.4, 8.5	—
—	Mobile app / portal access	N	Mobile / portal access not evidenced	5.11	—

Total features fully evidenced: 8 / 32 (a further 7 are partially evidenced; Y = implemented and evidenced, Partial = partly evidenced, N = not implemented, Not verified = not visible in the supplied evidence)

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	3	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	3	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	3	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Registration Enquiry Form → On Success → Deluge script (DS-01)

```
// Script ID : DS-01
// Location  : Registration Enquiry Form → Created → Successful form
submission
// Purpose   : Check Class XII percentage range and log values
// NOTE: only the lines visible in the supplied screenshot are
reproduced; paste the remaining lines
if(input.Class XII Percentage >= 0 && input.Class XII Percentage <=
100)
{
    info "Percentage validation successful";
}
else
{
    info "Invalid percentage";
}
info "Student Name: " + input.Student_Name1;
info "Email: " + input.Email_Address;
// line 11 onward not fully visible (the log shows Percentage and
Preferred Course are also printed)
```

C.2 Registration Enquiry Form → On Success → Deluge script (DS-02)

```
// Script ID : DS-02
// Location  : Registration Enquiry Form → Created → Successful form
submission
// Purpose   : Log that the workflow is running
info "WORKFLOW IS RUNNING";
```

C.3 Registration Enquiry Form → On Success → Deluge script (DS-03)

```
// Script ID : DS-03
// Location  : Registration Enquiry Form → Created → Successful form
submission
// Purpose   : Create a lead in Zoho CRM and log the response
// NOTE: line 7 is cut off at the screenshot edge; only visible lines
are shown
info "CRM ACTION STARTED";
leadInfo = Map();
leadInfo.put("Last_Name",input.Student_Name1);
leadInfo.put("Email",input.Email_Address);
leadInfo.put("Phone",input.Contact_Number);
leadInfo.put("Description",input.Preferred_Course);
response =
zoho.crm.v8.createRecord("Leads",leadInfo,Map(),"zoho_crm_connectic...
info "CRM RESPONSE:";
info response;
```

C.4 Further workflow script (DS-04)

```
// Script ID : DS-04
// Source code was not supplied. The log message "Matching enquiries:
1" indicates that such a script exists.
// [paste script here]
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application URL [insert URL] in a web browser and sign in with a Zoho account that has access to the application. The application opens with the left-hand menu described in Section 5.1. Mobile access was not evidenced.

D.2 Instructions for the Prospective Student

- Open **Admission Enquiries → Registration Enquiry Form**.
- Fill in the fields, starting with the required fields marked with an asterisk (Student_Name, Date_Of_Birth, Gender, Email_Address), then the remaining fields, and select **Submit**. A message "Admission Enquiry Added Successfully" confirms the submission.

D.3 Instructions for the Application Owner / Counsellor

- Open **Admission Enquiries → All Registration Enquiries** to view the submitted enquiries; use the search icon to look for a record.
- Open **Preferred Course** to view the pie chart, and **Course Registration Enquiry Dashboard** to view the gender chart, the web chart, the form and the report.